

The Millikan Oil-Drop Experiment: Finding the Elementary Charge

The Millikan Oil-Drop Experiment · oPhysics

Senior Secondary (Years 11-12)

~30 minutes

Science · Physics

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Simulation: <https://ophysics.com/em2.html>

Curriculum links: quantum physics · atomic structure

Learning intentions

- I can explain how balancing gravity and an electric force on a charged drop allows its charge to be determined.
- I can use repeated measurements of charge to find the elementary charge, e .
- I can explain the significance of finding that charge only occurs in whole-number multiples of e .

Getting started

Open the simulation and follow the on-screen instructions. Adjust the plate voltage until a chosen oil drop is suspended, motionless, between the plates.

Tasks

1. Record the mass of your first oil drop and the voltage needed to suspend it. Calculate the charge on the drop using the balance of gravitational and electric forces ($qE = mg$).

Working:

2. Repeat Task 1 for four more oil drops, recording the mass, voltage and calculated charge for each.

Drop mass

Suspending voltage

Calculated charge

3. Look at your five charge values from Tasks 1–2. Find the largest number that all five charges are (approximately) whole-number multiples of — this is your estimate of the elementary charge.
4. Compare your estimated elementary charge to the accepted value of $e = 1.60 \times 10^{-19}$ C. Calculate your percentage error.

Extension (optional)

Explain why the discovery that charge always comes in whole-number multiples of a smallest unit, e , was so significant for understanding the structure of matter.

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